OBINNA DOMINIC UKE

Applied Environmental Economist ◇ Climate, Energy, and Finance

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Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

FIELDS

Primary: Environmental and Energy Economics
Secondary: Financial Economics, Development Economics, Applied Econometrics

Research interests: Environmental and energy economics; carbon pricing and fossil fuel subsidies; carbon border adjustment and transition risk; climate risk and financial markets; global value chains and structural transformation; sovereign risk and credit markets; climate risk transmission and macroeconomic dynamics; climate finance and adaptation in the Global South.

EDUCATION

Ph.D. in Management and Organizational Sciences 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
Africa Business School, Mohammed VI Polytechnic University (UM6P), Rabat, Morocco

- Dissertation: *Climate Risk, Carbon Pricing, and Financial Markets: Essays in Environmental and Financial Economics*
- Supervisors: Prof. Amine Ben Amar (Finance Cluster, ABS, UM6P) and Prof. Olga Gorelkina (Moroccan Center for Game Theory, UM6P)

Joint Doctorate, Production Engineering (PEP/COPPE) 2023 to Aug. 2027 (expected)

Research field: Environmental and Energy Economics
COPPE, Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil

- Joint doctoral agreement with ABS, UM6P, conferring the doctoral degrees of both institutions on a single defence; co-supervisor Prof. Luan Santos

M.Sc. in Fertilizer Science and Technology Oct. 2020 to Jul. 2022
ESAFE, Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco

B.Sc. in Agriculture (Soil and Crop Sciences) Dec. 2013 to Aug. 2018
Usmanu Danfodiyo University, Sokoto, Nigeria

JOB MARKET PAPER

"The Carbon Footprint of Fossil Windfalls"
Sole authored. October 2026. JEL: Q32, Q54, E62, F41, H54, O13.

Every rise in the world price of oil, gas or coal transfers income to producer economies, and the energy transition will make these transfers more volatile before it makes them smaller. Using national accounts, government revenue and sector-level fossil CO_2 for 163 countries over 1981 to 2024, and a windfall shock built from predetermined fossil rents and unexpected world fuel prices, I estimate what windfalls do to producer economies and their emissions. In the 75 economies with resource-revenue data, almost all of the windfall reaches the government budget; most is saved abroad in the first year and absorbed into consumption, investment and imports within two. Each dollar of windfall revenue adds 0.52 kg of CO_2 at home (95% weak-instrument-robust interval: 0.20 to 0.89 kg), about half the carbon intensity of an average dollar of GDP, mostly through road transport,

power and buildings. Cement emissions trace an investment channel running through existing kilns. Instrumenting the windfall with OPEC oil supply news gives macroeconomic, construction, buildings and road transport responses of the same size, and an equally large rise in energy CO_2 in independent energy statistics. In cement, buildings and power, booms raise emissions and busts of the same size do not lower them. Over the 2003 to 2014 commodity boom, windfalls added about 2.7 billion tonnes of CO_2, valued at roughly 500 billion dollars at the social cost of carbon.

WORKING PAPERS

"Riding the Climate Wave: A Wavelet Analysis of Sovereign Risk in Developed and Developing Economies"

With Amine Ben Amar and Olga Gorelkina. Under review: International Review of Financial Analysis (FINANA-D-25-05094); revised version submitted September 2026. Presented: 2nd QFDAM, Amiens, June 2025; 71st EWGCFM, Rabat, May 2025.

We ask which dimension of climate risk sovereign credit markets price: physical exposure (vulnerability) or the capacity to adapt (readiness). Using 59 economies over 2008 to 2021, spanning 28 advanced economies and 31 emerging market and developing economies, wavelet coherence documents significant co-movement between climate vulnerability and sovereign credit default swap spreads at short, medium and low frequencies, neither concentrated in the long horizon nor systematically different between advanced economies and EMDEs. In the cross-section the vulnerability gradient vanishes once development is controlled, while readiness carries a strong negative association with spreads net of income, public debt and the Worldwide Governance Indicators: for EMDEs, closing the readiness disadvantage relative to advanced economies is associated with roughly 74% smaller spreads. It is adaptive capacity, not exposure, that the market rewards. We also document and correct a measurement convention that inflates conventional wavelet coherence summaries. Significance is assessed against an AR(1) red-noise null with 10,000 Monte Carlo simulations.

"Fiscal Capacity, Not Climate Exposure: Decomposing Global Fossil-Fuel Underpricing and Its Equity Burden"

With Olga Gorelkina and Amine Ben Amar. Submitted: Energy Policy. JEL: Q48, Q54, Q58, H23, H20, O23.

We construct a near-global panel of effective carbon prices for 174 countries from 2015 to 2022, integrating explicit carbon-pricing instruments with implicit fossil fuel subsidies against three social cost of carbon benchmarks, and decompose the underpricing shortfall into a domestic wedge, made up of supply-cost gaps, foregone consumption taxes and local air pollution, and a global climate wedge. Underpricing is near-universal: only Sweden, Norway and Finland price above zero. Fiscal capacity, measured by non-resource tax revenue, predicts the domestic wedge, which is roughly two thirds of the total, while no observable state characteristic predicts the climate wedge. Climate vulnerability does not raise deficits directly, yet vulnerability and weak fiscal capacity coincide so closely that the income-normalised burden is about twenty times larger in the most vulnerable quintile. Domestic revenue mobilisation is therefore climate policy for the larger part of the shortfall, while the climate wedge requires international coordination.

"Financial Reach and Household Coping under Climate Shocks: Evidence from Ten African Countries"

With Todd Cort (Yale School of Management). Submitted: World Development (WD-45053). JEL: Q54, O16, G51, D14.

Harmonised shock-and-coping modules from ten African countries permit a within-household comparison of responses to a covariate climate shock and to an idiosyncratic one, so household-fixed characteristics difference out. Covariate shocks are associated with a 1.9 to 2.2 percentage-point higher probability of reducing consumption, while idiosyncratic shocks more often trigger asset sales. In the 2018 West African sample a sender elsewhere in the country is associated with a 4.6 percentage-point smaller shift towards cutting consumption than a same-region sender, surviving administrative-region fixed effects, while credit access is statistically indistinguishable from zero. Growing-season rainfall

anomalies remain correlated across roughly 254 km, measured from a CHIRPS correlogram, which motivates but does not identify the reach measure. The estimates are associational. The contribution is conceptual as much as empirical: instrument ownership alone is an incomplete measure of climate resilience, because the exposure of the income backing a financial claim matters alongside the instrument a household holds.

"Fossil Fuel Underpricing and the Carbon Composition of Manufacturing Value Chains in Climate-Disaster Years"

Sole authored. Submitted: Economic Analysis and Policy. Replication package: 10.5281/zenodo.20468571. JEL: Q54, Q58, F18, L60, O14, H23.

Holding domestic energy prices below their efficient level can shield energy users from cost shocks, yet little is known about whether fossil fuel underpricing shapes how industrial structure adjusts to climate disasters. Combining global value chain output for 77 countries and 18 manufacturing sectors over 2016 to 2022 with IMF estimates of fossil fuel underpricing and EM-DAT disaster records, I find that in disaster years the share of manufacturing value-chain output produced by carbon-intensive sectors rises with underpricing ($p = 0.006$), by about one percentage point at the 90th percentile, while it falls slightly and insignificantly where fuel is priced close to cost. The pattern holds with country-specific trends and with a control for fossil rent exposure, and sector-level estimates with country-sector, country-year and sector-year fixed effects point the same way ($p = 0.041$). Sanctions and armed conflict show no comparable pattern, and the association is strongest in resource-rich economies with weak democratic accountability. Because underpricing varies mainly across countries, the estimates compare pricing regimes and are conditional associations. Appraisals of fossil fuel subsidy reform and of disaster relief should therefore consider how pricing regimes shape the sectoral incidence of climate shocks, alongside fiscal savings and emission reductions.

"Heat Beyond the Farm Gate: Offsetting Supply and Demand Responses in European Food and Drink Manufacturing"

With Amine Ben Amar and Olga Gorelkina. October 2026. JEL: C23, L66, Q11, Q13, Q54.

Heat reaches food and drink manufacturers through the supply of farm inputs and the demand for their products, and the two channels can move output in opposite directions. We estimate monthly responses to weather for food and drink industries in 21 European countries, and for raw milk deliveries and slaughter, over 2000 to 2026, identifying responses from differences between countries' weather in the same month with panel local projections. In the average country a summer month 1°C warmer than normal lowers raw milk deliveries by 0.3% and bakery output by 0.8% and raises beverage output by 2.0%. The milk and meat losses follow days above 30°C rather than heat that is unusual for the season, occur in warm but not cool months and are not recovered, as livestock heat stress would produce; beverage output follows both kinds of heat and moves with the price of non-alcoholic beverages, as a demand shift would. Because the responses offset, 81% of the gross change in value added cancels, the food products total barely moves, and beverages are the only one of 24 manufacturing divisions to respond. Aggregate food output therefore understates how exposed the livestock-based food chain is to heat.

"Building Resilience, Reducing Risk: A Systematic Review of Climate Resilience, Climate Finance Impacts, and Market Mispricing in the Global South"

With Amine Ben Amar and Olga Gorelkina. Retained as dissertation material.

A PRISMA-based systematic literature review consolidating the relationship between climate resilience initiatives, financial risk reduction and market mispricing in the Global South. Firms in climate-vulnerable regions face higher borrowing costs and banks impose stricter loan terms in high-risk geographies. Effective governance and precautionary financial policies are identified as critical moderators of climate finance impact.

WORK IN PROGRESS

"Carbon Accounting Capability and Exposure to the EU Carbon Border Adjustment Mechanism: Evidence from African Exporters"

Target: Business Strategy and the Environment.

Border carbon pricing is usually studied as trade policy falling on countries. It falls on consignments, and charges a conservative default unless a producer verifies actual emissions inside the regulatory boundary. The capability it rewards is boundary fit: emissions measured on the system boundary the regulating jurisdiction prices. I compute liability line by line for 1,760 eight-digit tariff codes from European import declarations and official default values, deducting the year-specific free-allocation credit along the statutory phase-in path. African exporters face EUR 1,768.1 million annually on the 2023 to 2025 mean at full phase-in, with incidence reaching 49.0% of customs value on semi-finished steel against a zero applied tariff, so the mechanism reintroduces the tariff escalation that trade policy removed. Five installation cases show boundary fit distributed independently of ownership form, firm size and country income: the best-placed producer is 94% state-owned, while a listed multinational's subsidiary publishes nothing at installation level. Because the regulation credits only explicitly paid carbon prices, 24 to 69% of the required reform is invisible to the deduction.

"Climate Shocks and Food Manufacturing: Aggregation Bias in the United Kingdom"

With Amine Ben Amar and Olga Gorelkina. Presented: 3rd QFDAM, Rabat, June 2026.

Climate-macro studies typically model the food system as a single sector. Using UK quarterly data for 1990 to 2025, we separate primary agriculture from downstream food manufacturing and ask what a single food-sector climate coefficient conceals. A food-system VARX estimated on the same data has an impact response less than half the agricultural one, and a Monte Carlo experiment shows that it systematically erases the persistent decline the disaggregated estimates imply. A single food-sector climate coefficient therefore depends on sectoral composition and is not a structural parameter. The sectoral response estimates are being rebuilt.

"Sovereign Climate Risk and Corporate Bond Spreads: Evidence from Emerging Markets"

With Luan Santos (COPPE, UFRJ).

Monthly panel of emerging market corporate bonds from 2018 to 2024, matched to sovereign credit default swaps, rating actions, EM-DAT disasters and physically measured tropical cyclone tracks, with pre-registered hypotheses and randomisation inference. Current estimates find no sovereign transmission, no change in market access and no average corporate repricing after disasters, and do not support state backing as an insulation mechanism. Developed under the joint doctoral agreement with COPPE, UFRJ.

PEER-REVIEWED PUBLICATIONS

Guéablé, Y.K.D., **Uke, O.D.** et al. (2022). Study of the Sodicity of Phosphate By-Products and Sludge Mixture for Large-Scale Application in Mine Site Reclamation. *Environmental Sciences Proceedings*, 16(1), 18.

Uke, O.D. & Haliru, M. (2021). Salinity Study of the Soils of Fadama Farms, Sokoto, Nigeria. *Farming and Management*, 6, 1-7.

Yusuf, Y.M., Bagudo, H.A. & **Uke, O.D.** (2021). Effects of Weed Control on Growth and Yield of Pearl Millet in the Sudan Savanna, Sokoto State, Nigeria. *Journal of Current Opinion in Crop Science*, 2(2), 155-164.

CONFERENCE PRESENTATIONS

3rd Quantitative Finance and Data Analytics Meeting (QFDAM 2026), Rabat, Morocco June 2026

Climate Shocks, Sectoral Transmission, and Monetary Policy: Evidence from UK Food Manufacturing

2nd Quantitative Finance and Data Analytics Meeting (QFDAM 2025), Amiens, France June 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

71st EURO Working Group for Commodities and Financial Modelling (EWGCFM 2025), Rabat, Morocco May 2025
Riding the Climate Wave: Wavelet Analysis of Sovereign Risk in Developed and Developing Economies

TEACHING

Teaching Assistant, Financial Econometrics [Master 2] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Microeconomics [Licence 1] Sep. to Nov. 2026
Mohammed VI Polytechnic University (UM6P) (Instructor: Prof. Amine Ben Amar)

Teaching Assistant, Essentials of Financial Markets [M1] Jan. to Feb. 2025
Africa Business School, UM6P (Instructor: Prof. Amine Ben Amar)

PROFESSIONAL EXPERIENCE

PhD Researcher, Environmental and Energy Economics Oct. 2023 to Present
Africa Business School, UM6P, Morocco; COPPE, UFRJ, Brazil

Produce policy-relevant research on climate shock propagation, physical risk pricing, carbon pricing failures and sovereign debt vulnerability. Build and validate multi-source climate, trade, firm and household datasets, and publish reproducible replication packages.

Regional Field Project Manager Jul. 2022 to Oct. 2023
OCP Africa Nigeria

Coordinated multi-regional agricultural development programmes across Southeast and South-South Nigeria in partnership with USAID/TradeHub. Managed programme budgets, donor reporting and workplan tracking. Produced policy briefs on climate-smart agriculture for government counterparts and programme leadership.

Research Agronomist / Blending Plant Support Jun. 2021 to Jun. 2022
OCP Africa Nigeria

Designed and executed fertilizer validation field trials; prepared grant applications and supported monitoring and evaluation for USAID-funded agricultural initiatives.

Founder and Chief Technical Officer Dec. 2020 to Jan. 2022
AgriBox SARL, Morocco

Founded a climate-smart hydroponic enterprise; developed impact investment proposals and private sector partnerships.

Project Manager, Microfinance Operations Jun. 2019 to Mar. 2020
Bincom ICT Solutions (Social Lender), Nigeria

Managed microfinance loan disbursement, risk assessment and financial proposals across Kano State. Served as institutional liaison between Social Lender and the banking sector.

FELLOWSHIPS AND AWARDS

Full Doctoral Merit Scholarship, Africa Business School, UM6P, Morocco	2023
HULT Prize in Innovation, Mohammed VI Polytechnic University	2021
NARSAD Deliberative Polling Award, UM6P and Stanford University	2021

IBN Rochid Foundation Scholarship, M.Sc. Studies, OCP Group 2020

TECHNICAL SKILLS AND LANGUAGES

Statistical software: R, Stata, Python, SQL, GIS, VS Code, Git, Advanced Excel.

Methods: SVAR and local projections, including time-varying local projections with stochastic volatility; wavelet and partial wavelet coherence; triple fixed effects with cluster-robust inference; within-household first differences; event study designs with continuous exposure; IV and weak-instrument diagnostics; leave-one-out robustness; PRISMA systematic review.

Data and geospatial: construction and validation of multi-source country, sector, firm and household panels; CHIRPS, CHIRTS, ERA5 and Sentinel-1 products; Eurostat Comext and UN Comtrade at tariff-line level; World Bank LSMS-ISA and EHCVM household surveys; IMF fossil fuel subsidy and carbon pricing data; reproducible pipelines with replication packages deposited on Zenodo.

Languages: English (native), Igbo (native), Portuguese (intermediate), French (basic).